Bing Li

(+65) 80674730 | bing.li@ntu.edu.sg | https://libingsy.github.io/

Research Fellow | Nanyang Technological University

Education

Beihang University (Ph.D.) **Sep 2018 – Jun 2024**

Computer Application Technology (Advisor: Prof. Di Huang) Beijing, China

Northeastern University (Master) **Sep 2013 – Jun 2015**

Signal and Information Processing Shenyang, Liaoning Province, China

Northeastern University (Bachelor) **Sep 2009 – Jun 2013**

Electronic Information Engineering Shenyang, Liaoning Province, China

Employment

Nanyang Technological University, Singapore **Jan 2026 – Present**

Research Fellow (Advisor: Prof. Wee Peng Tay) Research on UAV perception systems

CFAR, A*STAR, Singapore **Oct 2024 – Oct 2025**

Postdoctoral Researcher (Advisor: Prof. Qing Guo) Research in Adversarial Attacks for Autonomous Driving Systems

China Mobile, Shenyang, China **Jun 2015 – Sep 2018**

Staff Engineer Network infrastructure development and maintenance

Research Interests

- UAV Perception, Multi-Modal 3D Perception, Autonomous Driving Safety
- Computer Vision, Image/Video Understanding
- 3D Generation, Adversarial Attacks, Gaussian Splatting

Publications

Published / Accepted

[1] **Bing Li**, Wuqi Wang, Yanan Zhang, Jingzheng Li, Haigen Min, Wei Feng, Xingyu Zhao, Jie Zhang, Qing Guo. OBJVanish: Prompt-Driven Generation of Physically Realizable 3D LiDAR-Invisible Objects. (ICML 2026).

[2] **Bing Li**, Sean Du, Xuhong Ren, Luqi Gong, Wee Peng Tay, Qing Guo. Agent-OBJ: Prompt-Driven 3D Adversaries for Multi-Modal Perception. (ECCV 2026).

[3] **Bing Li**, Guohao Li, Jiaxin Chen, Dongming Zhang, Xiuguo Bao, Di Huang. Cross-Modal Contrastive MAE for Compressed Video Pre-Training. (IEEE Transactions on Image Processing 2025).

[4] **Bing Li**, Jiaxin Chen, Xiuguo Bao, Di Huang. Compressed Video Prompt Tuning. (NeurIPS 2023).

[5] **Bing Li**, Jiaxin Chen, Dongming Zhang, Xiuguo Bao, Di Huang. Representation Learning for Compressed Video Action Recognition via Attentive Cross-Modal Interaction with Motion Enhancement. (IJCAI 2022).

[6] **Bing Li**, Longteng Kong, Dongming Zhang, Xiuguo Bao, Di Huang, Yunhong Wang. Towards Practical Compressed Video Action Recognition: A Temporal Enhanced Multi-Stream Network. (ICPR 2021).

[7] Hong Wen[1], **Bing Li**[1], Liang Qian, Haoyuan Li, Weiwei Fang. EMM-CVT: An Efficient Multi-Modal Video Transformer for Compressed Video Action Recognition. (Pattern Recognition 2026).

[8] Yuwei Ding, **Bing Li***, Yian Zhu, Yanan Zhang, Bohao Qu, Xueping Wang, Qing Guo. ReAlignHDR: Region-Constrained Alignment for Ghost-Free HDR Imaging. (Pattern Recognition 2026).

[9] Zhewen Zhou, **Bing Li***, Qing Guo, Chenlei Li, Guangshuai Gao. PACTFormer: Peak-Aware Cross-Temporal Transformer for Temporal Action Detection. (PRCV 2025).

[10] Jingzheng Li, Xianglong Liu, Shikui Wei, Yufei Ge, Zhijun Chen, **Bing Li**, Qing Guo, Xianqi Yang, Yanjun Pu, Qianren Mao, Jiakai Wang. Toward Benchmarking and Assessing the Safety and Robustness of Autonomous Driving on Safety-Critical Scenarios. (IEEE Transactions on Image Processing 2026).

[11] Bohao Qu, Xiaofeng Cao, **Bing Li**, Menglin Zhang, Tuan-Anh Vu, Di Lin, Qing Guo. Exploiting Geometric Structures for Modeling Multi-Agent Behaviors: A New Thinking. (AAAI 2026).

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* Corresponding author; [1] Co-first author.

[12] Xueping Wang, Xueni Guo, Feihu Yan, **Bing Li**, Guangzhe Zhao. Progressive Densification of 3D Gaussians for High-Fidelity Talking Head Synthesis. (Frontiers of Computer Science 2026).

[13] Dongmei Chen, Guiji Sun, **Bing Li**, Shuhua Liu, Jv Luo, Xianfeng Zhou, Jingcheng Zhang. TSHFS: A Two-Stage Hyperspectral Feature Selection Approach for Congeneric Rice Planthopper Species Classification. (Pest Management Science 2026).

Under Review

[1] **Bing Li**, Jiaxin Chen, Xiuguo Bao, Di Huang. CVTracker: Learning to Track Objects in Compressed Videos. (Pattern Recognition 2026 submitted).

[2] **Bing Li**, Yansen Ma, Ruikui Wang, Zhijun Chen, Yuwei Ding, Sean Du, Jinjin Zhang. LOST: Localized Object-Level Semantic-Field Tampering in Open-Vocabulary 3DGS. (AAAI 2027 submitted).

[3] Qinghua Zhang, **Bing Li***, Shuyu Hu, Yanan Zhang, Bohao Qu, Zhijun Chen, Yunpeng Liu. FSG-Net: Foundation Semantic Guidance for Region-Consistent Low-Light Image Enhancement. (AAAI 2027 submitted).

[4] Qinghua Zhang, **Bing Li***, Tong Jin, Haiguang Wang, Jingzheng Li, Sean Du, Wuqi Wang, Yunpeng Liu. RefSift-GS: Reference Sifting in 3D Gaussian Splatting for Novel View Synthesis under Challenging Illumination. (AAAI 2027 submitted).

[5] Mengkun Guo, **Bing Li***, Ming Ma, Xiaoyang Hao, Bohao Qu, Jingzheng Li, Wuqi Wang, Zhijun Chen. TextComp: Text Guided Representation Learning for Compressed Video Action Recognition. (AAAI 2027 submitted).

[6] Yuwei Ding, **Bing Li***, Yanan Zhang, Bohao Qu, Xueping Wang, Guangshuai Gao, Sean Du. From Filtering to Routing: Prediction-Interval-Guided Semi-Supervised HDR Deghosting. (AAAI 2027 submitted).

[7] Yansen Ma, **Bing Li***, Xianjing Han, Xueyang Kang, Bohao Qu, Qing Guo, Wee Peng Tay. LAWS: LiDAR-Anchored Multimodal Weather Synthesis with Physically Grounded 4D Dynamics. (AAAI 2027 submitted).

[8] Xinqi Li, Liming Bao, **Bing Li***, Ze Li, Zhijun Chen, Bohao Qu, Qing Guo, Tao Sun. SIDRAG: Adaptive Defense against RAG Poisoning via Set-Level Influence Detection. (ICLR 2027 submitted).

[9] Bohao Qu, Yu Wang, Hongwei Wang, **Bing Li**, Wanbing Zhang, Fen Fang, Xiuju Fu, Qing Guo. GeoDiffusion: Spatio-Temporal Offset-Aware Latent Diffusion for Irregular Trajectories. (ICLR 2027 submitted).

Projects

Deep hash video retrieval **Oct 2019 – Jan 2021**
hybrid GPU/CPU decoding

Classification and annotation of video complex scene datasets **Nov 2018 – Jun 2020**
dataset construction and automated labeling

Skills

Research: multi-modal 3D perception (LiDAR/camera), 3D Gaussian Splatting, diffusion models, adversarial attacks and defenses, compressed video analysis
Programming: Python, PyTorch, C/C++, Linux, LaTeX

Academic Services

Reviewer of IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) and IEEE Transactions on Multimedia (TMM)
Reviewer for international conferences (NeurIPS, IJCAI, AAAI, MM, CVPR, and ECCV)